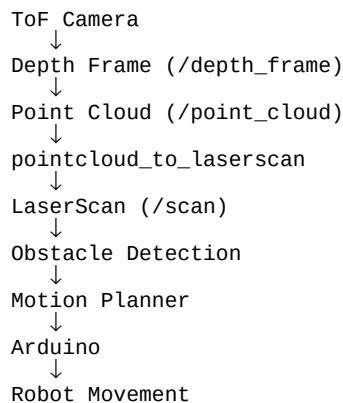


Eon Navigation Notes (Detailed)

These notes capture the concepts we discussed, starting from 'How do I use the ToF camera for navigation?' They are written as a revision guide.

1. Overall Pipeline



2. What is the Depth Frame?

The depth frame is an image where every pixel stores a distance instead of a color. The values are usually in millimetres before your code converts them to metres.

Many values become 0 because your code removes unreliable measurements using confidence filtering:

```
depth_buf[confidence_buf < 30] = 0
```

A zero therefore means 'invalid or unreliable measurement', not necessarily that an object is touching the camera.

3. Point Cloud Recap

Every pixel knows two things: (1) where it is in the camera image (its x,y pixel position) and (2) the measured depth. Using the camera calibration (fx, fy), the pixel is converted into a real 3D coordinate.

```
Pixel + Depth
  ↓
(x, y, z)

x = left/right
y = up/down
z = distance from camera
```

The camera does NOT emit one infrared beam per pixel. An infrared light source illuminates the scene and every camera pixel observes the reflected light coming from a slightly different direction. Because each pixel looks in a different direction, each pixel measures a different point in space.

4. Why do we need x and y?

If only z (distance) existed, the robot would know something is 2 metres away but not whether it is to the left, right, up or down. x and y tell where the point is located, while z tells how far away it is.

5. How can a 3D point exist if the light travels straight?

Each pixel looks along its own straight viewing direction. The viewing directions are spread across the camera's field of view. Every pixel therefore measures a different location in the scene. Combining all pixels creates a cloud of 3D points.

6. What is LaserScan?

The simplest definition, and the one that finally made the concept click:

LaserScan = Angle → Nearest Obstacle Distance

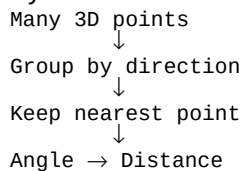
Conceptually:

Angle	Distance
-90°	2.4 m
-60°	2.2 m
-30°	1.8 m
0°	0.45 m
+30°	1.6 m
+60°	2.3 m
+90°	2.4 m

ROS actually stores angle_min, angle_max and angle_increment plus an array called ranges. The angle for each element is computed from those values.

7. Point Cloud to LaserScan

The converter examines one direction at a time. In each direction there may be many 3D points. It keeps only the closest one because that is the first obstacle the robot would hit.

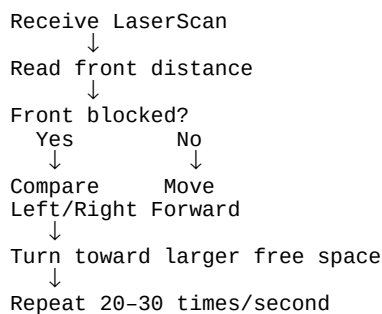


The wall behind a chair is ignored because the chair is the first obstacle.

8. Why use LaserScan?

Navigation does not usually need tens of thousands of 3D points. It mainly needs to know how far the nearest obstacle is in every direction. LaserScan is therefore much simpler and many ROS navigation algorithms expect it.

9. Simple Navigation Algorithm



10. Future Plan for Eon

First convert the ToF point cloud into a LaserScan. Build a lightweight local obstacle avoidance system. Later add occupancy grids, mapping, SLAM or Nav2 if needed. This keeps the robot simple while leaving room for future upgrades.